

Call Center Performance Analysis

End-to-End Case Study — Python · SQL · Power BI

Author: Franklin Manuel Ruiz Guadamuz · Tools: Python (pandas), SQLite, Power BI · Source: Kaggle — Call Center Data (1,251 records)

1. PROJECT OVERVIEW

This project analyzes 1,251 contact-center interaction records end to end, using Python for cleaning, SQL for analysis, and Power BI for visualization. It answers how consistently the operation meets its service-level (SL) target, and what drives SL breaches and abandonment. The angle draws on 8+ years of Workforce Management experience: the interpretation — not just the charts — is what turns the metrics into an operational diagnosis.

2. DATASET

Attribute	Detail
Source	Kaggle — Call Center Data
Rows	1,251 interaction records
Metrics	incoming / answered / abandoned calls, answer rate, answer speed (ASA), talk duration (AHT), waiting time, service level (20s)
Assumption	80% SL target (industry standard; not specified in the data — documented as an assumption)

3. PIPELINE & TOOLS

Python (pandas) cleans the raw CSV, then loads it into **SQLite** programmatically via the `sqlite3` library. **SQL** runs the analysis on the `calls` table, and **Power BI** connects to the same database over ODBC to build the dashboard. A useful integrity check: SL compliance comes out at 36.6% in Python, SQL, and Power BI alike — confirming the pipeline is consistent across all three tools.

4. DATA CLEANING & FEATURE ENGINEERING (PYTHON)

The raw data stored several metrics as text. Cleaning steps:

- **Percentages to numbers:** stripped the % from answer rate and service level and cast to float.
- **Durations to seconds:** converted "0:02:14" strings with `pd.to_timedelta(...).dt.total_seconds()`.
- **Derived metrics:** abandon rate (`abandoned ÷ incoming × 100`) and a `sla_met` flag (`service level ≥ 80`).
- **Standardized column names** to `snake_case`, then loaded to SQLite.

```
# Durations to seconds
df['talk_duration_avg'] = pd.to_timedelta(df['Talk Duration (AVG)']).dt.total_seconds()

# Derived metrics
df['abandon_rate'] = df['Abandoned Calls'] / df['Incoming Calls'] * 100
df['sla_met']      = df['Service Level (20 Seconds)'] >= 80 # 80% target

# Load to SQLite from Python
import sqlite3
conn = sqlite3.connect('callcenter.db')
df.to_sql('calls', conn, if_exists='replace', index=False)
```

5. ANALYSIS & FINDINGS (SQL)

Overall snapshot

Across 1,251 records: SL compliance **36.6%** (the 80% target is met barely a third of the time), average service level **70.9%**, average abandon rate **7.3%**, average AHT **158 s**.

Volume vs service level

Volume	Records	Avg SL	Abandon	Compliance
Low (<100)	253	77.9%	5.0%	54.2%
Medium (100-250)	734	72.6%	6.1%	34.6%

High (250-500)	214	65.8%	9.2%	29.9%
Very High (>500)	50	31.8%	28.8%	6.0%

Finding: service level falls steadily as volume rises and abandonment climbs — at Very High volume, SL collapses to 31.8% and nearly one in three callers abandons. Critically, even the Medium bucket (the bulk of records) sits below the 80% target, which points to a **base staffing gap, not just a peak-handling problem**.

Wait time vs abandonment

Wait time	Records	Abandon	Avg SL
Under 1 min	77	1.7%	93.3%
1-3 min	544	3.9%	79.0%
3-5 min	359	7.0%	68.1%
Over 5 min	271	16.2%	51.9%

Finding: abandonment multiplies roughly 10x as wait grows, and SL drops in parallel. The threshold is telling — once wait exceeds ~1 minute, SL already falls below the 80% target, giving a concrete, actionable limit for acceptable wait.

Efficiency (AHT / ASA)

Average AHT 158 s (range 57-288 s) and average ASA 25 s (max 308 s). Handle times are stable and healthy — no runaway calls.

6. KEY INSIGHT

The metrics form a single causal chain: **higher volume → longer waits → more abandonment → lower service level**. Because AHT stays healthy throughout, the root cause is **staffing/dimensioning, not agent productivity** — the operation simply doesn't have enough capacity for its volume, and misses the SL target even at normal load. This is a WFM-informed diagnosis: the fix is capacity planning, not squeezing agents.

7. DASHBOARD (POWER BI)

A dark, professional dashboard (custom teal theme) connected to SQLite over ODBC. It leads with KPI cards, pairs the two causal stories (volume→SL and wait→abandonment) as contrasting bar charts, and closes with the compliance split and a volume-vs-abandonment scatter. Custom report-page **tooltips** surface per-bucket detail on hover.

Call Center Performance Analysis

1,251 interaction records · service level, volume & abandonment

36,61

SL Compliance %

70,87

Avg Service Level %

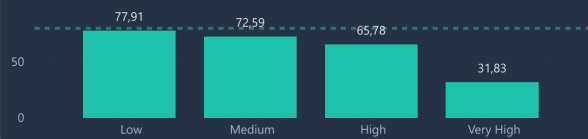
7,31

Avg Abandon Rate %

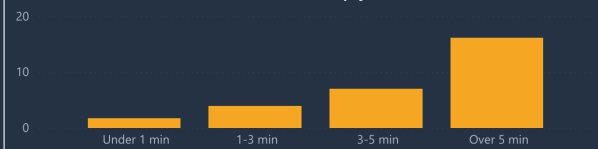
248,37K

Total Incoming

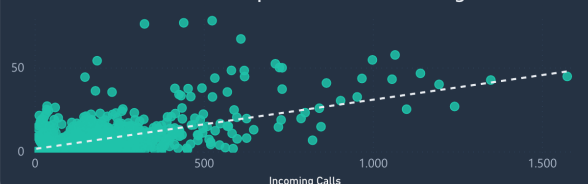
Service level collapses as volume rises



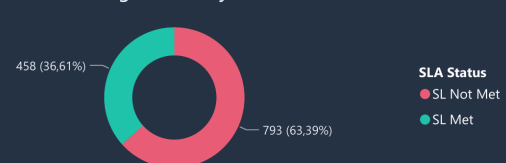
Abandonment climbs sharply with wait time



Abandonment spikes when volume surges



SL target met only 37% of the time



Call Center Performance Analysis — Power BI dashboard (SQLite via ODBC, custom tooltips)

8. LIMITATIONS

The dataset has no timestamps, agent IDs, or queue/LOB fields, so time-of-day and day-of-week staffing patterns, forecast-vs-actual accuracy, and per-agent or per-queue analysis are out of scope. The 80% SL target is an assumed industry standard. The 'Very High' volume bucket rests on 50 observations, so that figure is directional.

Pipeline: Python (pandas cleaning) → SQLite (loaded via sqlite3) → SQL analysis → Power BI (ODBC) with custom tooltips. **Consistency check:** SL compliance = 36.6% in all three tools.